

transistors that can be turned ‘on’ or ‘off’, and always exist in a single state. In classical computation, the unit of information is a bit, which can be 0 or 1. All processing can be done by simple logic gates (AND, NOT) acting on these bits. The states are distinguishable and observed without disturbing the system.

Quantum computers use ‘quantum bits’, or ‘qubits’, which are a superposition (that can exist in more than one state) of 0 and 1. A 2-qubit quantum machine allows you to do four calculations at once. A 3-qubit quantum machine can do eight calculations. A 4-qubit machine gives you 16 calculations, simultaneously. A quantum machine with ‘n’ qubits can be in a superposition of up to 2^n unique states simultaneously. The state of a qubit can depend on the state of another qubit. Due to this, the qubits can be entangled, which is another characteristic of quantum mechanics.

This capability enables quantum computers to solve certain types of complex problems and do parallel processing, which is not possible with conventional computers. The power of qubits is that they can exponentially scale and speed up, as the computations occur in parallel.

Types of quantum computers

The various types of quantum computers in use today are:

- Superconductor-based quantum computers (including SQUID-based quantum computers)
- Ion trap based quantum computers
- Quantum computers based on ‘nuclear magnetic resonance on molecules in solution’
- Gate model quantum computers
- Quantum computers based on ‘quantum dot on surface’
- Quantum computers based on ‘laser acting on floating ions (in vacuum)’
- Molecular magnet based quantum computers
- Solid state NMR Kane quantum computers
- Adiabatic quantum computers

There are a few different ways to create a qubit. One method uses superconductivity to create and maintain a quantum state. To work with these superconducting qubits for extended periods of time, they must be kept very cold. Any heat in the system can introduce errors, which is why quantum computers operate at temperatures close to absolute zero.

The quantum computing technology landscape

There are many leading organisations working on quantum computing (QC). IBM, Microsoft, Alibaba, D-Wave Systems, Google, Nokia, Fujitsu and Intel continue to work and compete with each other in this field. As the research in QC matures, these companies are working rigorously to deliver seamless data encryption, real conversations with AI, and better financial modelling through quantum technology.

IBM: IBM has been working to develop a quantum computer for over 35 years. Less than a year ago, the company released a 5-qubit and a 20-qubit quantum computer, which offered quantum computing as an advanced cloud service. Multinational companies such as JPMorgan Chase, Daimler (which uses QC in logistics for optimising vehicle delivery routes), Honda, Samsung and Barclays (which uses QC for the settlement of large batches of transactions) were the first to sign up for the active testing of the service.

Google: Google has announced a 72-qubit quantum processor called Bristlecone, and signed an agreement with NASA to achieve supremacy in this field.

Microsoft: Microsoft has announced the Microsoft Quantum Network, which is a formal coalition of developers, partners, universities, leading startup talent, quantum innovators and customers to share knowledge and build a quantum computer, collectively. The coalition has open sourced the quantum computing development kit, which includes the quantum programming language Q# and the simulator.

Microsoft and Alphabet have partnered with Brilliant to introduce an online curriculum on quantum computing. It covers the basic concepts and introduces Microsoft’s Q# language with simple algorithms, before moving into complex and advanced scenarios.

D-Wave Systems: D-Wave has built the world’s first commercially available quantum computer with a simpler technology but with limited capabilities. The research focus of D-Wave is on how quantum computing can be applied to artificial intelligence, machine learning and difficult optimisation problems. In January 2017, the company sold its first 2000 qubit D-Wave 2000Q to cyber security firm Temporal Defense Systems.

Intel: In January 2018, Intel announced the delivery of a 49-qubit test quantum processor called Tangle Lake.

IonQ: IonQ is building a quantum computer based on the trapped ion technology.

Alibaba: Alibaba has designed an experimental quantum computer that is available online. In March 2018, it launched its ‘superconducting quantum computing cloud’ to provide access to an 11-qubit quantum computer. Developed with the Chinese Academy of Sciences, this allows users to run quantum programs and download the results.

Quantum computing applications and users

The implications of true quantum computing at scale are staggering and can have a tremendous impact on society. Quantum algorithms can be applied in a variety of industries for the following:

- Optimisation problems such as scheduling and route planning — to find out the best possible options out of a large number that are available
- Search, sampling and pattern matching
- Quantum encryption
- Industries – life sciences, finance, logistics, robotics and more

Machine learning and computer science: QC processes orders of huge magnitude more effectively than classical computing can. It allows the comparison